# Amara Okonkwo

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## EDUCATION

**University of Cambridge** | *MPhil Computer Science (Distributed Systems & Networks)* 2017–2019

- Achieved Distinction (3.92 GPA equivalent); thesis on low-latency trace aggregation for microservices reduced collection overhead by 34%
- Led Systems Research Society (40+ members), organized 8 seminars on observability at scale featuring speakers from Stripe and Google

**University of Lagos** | *BSc Computer Science* 2013–2017

- First Class Honours (3.89 GPA); awarded Best Computer Science Graduate; competed in ACM ICPC West Africa regionals
- Developed distributed fault-tolerant caching system as capstone; published findings in university tech journal with 120+ citations

## EXPERIENCE

**Datadog** | *Senior Platform Engineer, Observability Infrastructure* 2022–Present

- Led redesign of trace ingestion pipeline handling 2.1M spans/sec globally; reduced P99 latency from 850ms to 120ms (86% improvement) and cut infrastructure costs by $4.2M annually across 12-region deployment
- Owned distributed cardinality control system for metrics; prevented 7 customer-impacting outages by implementing real-time tag explosion detection, improving platform stability to 99.995% SLA (reduced incidents by 68%)
- Mentored 6 junior engineers on observability fundamentals; drove 3 RFCs through architecture review process for agent optimization, achieving 28% reduction in agent memory footprint (45MB → 32MB baseline)

**Cloudflare** | *Platform Engineer, Reliability & Telemetry* 2020–2022

- Architected distributed tracing system for 200+ edge locations processing 180K requests/sec; achieved 98.2% trace capture rate with <2% latency overhead via intelligent sampling and context propagation optimization
- Built anomaly detection pipeline for DNS query patterns; flagged security threats with 94% precision, preventing 12+ DDoS incidents affecting 50M+ users across $2.1B customer base
- Improved observability cost efficiency by implementing adaptive sampling and compression; reduced telemetry egress by 62% while maintaining 99.1% coverage of critical error paths

**Stripe** | *Software Engineer, Platform Infrastructure* 2019–2020

- Designed and deployed distributed tracing instrumentation across 4 core payment processing services; reduced MTTR by 41% (from 23 min to 13.5 min) and enabled root cause analysis of 99.2% of P1 incidents within 3 minutes
- Optimized metrics cardinality management; identified and eliminated 15.2M unused metric combinations, reducing Prometheus storage by 38% and monthly cloud spend by $180K
- Established observability SLO framework for 50+ service teams; achieved <4% SLO violation rate across 8 critical business flows handling $50B+ annual transaction volume

**Wise** | *Junior Platform Engineer* 2018–2019

- Implemented first unified logging pipeline for 8 core services; aggregated 1.8B log entries/day with sub-100ms query latency, enabling faster incident triage and reducing support tickets by 24%
- Contributed to on-call tooling improvements; shipped dashboards that reduced paging latency by 12s average, and built automation to resolve 31% of low-severity alerts autonomously
- Developed cost optimization analysis tooling that identified $127K in wasted infrastructure spend quarterly; documented findings that informed 3 major architectural decisions

## PROJECTS

**MetricFlow** | *Go, Prometheus, eBPF, PostgreSQL* 2021–Present

- Open-source distributed metrics cardinality manager with 2,847 GitHub stars; used by 340+ organizations to prevent cardinality explosions in Prometheus; achieved 4.2M NPM/PyPI downloads YTD
- Reduces metrics cardinality by avg 56% via intelligent tag dropping and aggregation; eBPF kernel module provides <0.1% CPU overhead for inline classification on 1M+ active metric streams

**TraceAI** | *Rust, Apache Arrow, ML (scikit-learn), Docker* 2020–Present

- Anomaly detection service for distributed traces with 1,156 GitHub stars; processes 900K traces/hour with 89ms P99 detection latency; 92% precision on root cause candidate ranking
- Reduced false positive rate from 18% (baseline heuristics) to 3.2% via supervised ML on trace feature extraction; community contributions yielded 8 high-impact PRs from external developers

**ObservabilityLens** | *Python, FastAPI, TypeScript/React, Grafana* 2019–2021

- Educational platform for teaching SRE observability concepts; 847 GitHub stars; hosted workshops at 6 conferences (KubeCon NA 2020, Velocity London 2021, DevOps Days Berlin 2021); trained 2,100+ engineers
- Built cost simulator module that helped 340 users estimate observability spend; reduced median implementation time for telemetry strategies by 34% (9.2 days → 6.1 days)

## EXTRACURRICULAR

**KubeCon + CloudNativeCon** | *Speaker & Track Chair, Observability* 2021–Present

- Delivered keynote 'Scaling Traces to Billions: Lessons from Production' at KubeCon North America 2022 (1,200+ attendees); received 4.7/5 feedback score with 340+ post-talk discussions
- Curated 16-session observability track for KubeCon Europe 2023; selected speakers from 140+ submissions; track achieved highest attendance (2,100+ engineers) across 3 sub-tracks

**Cloud Native Computing Foundation (CNCF)** | *Open Source Maintainer & Mentor* 2020–Present

- Maintain OpenTelemetry Go SDK; 2.1M+ weekly downloads; review 60+ PRs monthly and triaged 480+ GitHub issues YTD; onboarded 12 new contributors with structured mentorship leading to 14 merged contributions
- Mentor 4 junior engineers on observability architecture through CNCF mentorship program; 3 mentees promoted to senior roles and 1 published technical blog post on distributed tracing that reached 18K readers

## TECHNICAL SKILLS

**Languages** – Go, Python, Rust, TypeScript, SQL, Java, Bash

**Technologies** – Prometheus, Grafana, Elasticsearch, Jaeger, Datadog, OpenTelemetry, Kubernetes, Docker, gRPC, Apache Kafka, PostgreSQL, Redis, eBPF, Apache Arrow

**Tools** – Git, GitHub, ArgoCD, Terraform, CI/CD (GitHub Actions, GitLab CI), PagerDuty, Honeycomb, Linux profiling (perf, flamegraphs), Jupyter, DuckDB